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Received: 30 September 2018 | Revised: 6 December 2018 | Accepted: 20 December 2018

DOI: 10.1002/xrs.3017

SPECIAL ISSUE ARTICLE

WILEY

# ARDESIA: A fast silicon drift detector X-ray spectrometer for synchrotron applications

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**Funding information**
INFN

ARDESIA, a four-channel X-ray spectrometer based on silicon drift detectors (SDDs), is presented. It has been developed for synchrotron applications targeting especially X-ray fluorescence (XRF) and X-ray absorption spectroscopy (XAS) with good energy resolution at high count rates (a few Mcps per second). The main target applications are XRF and XAFS techniques. The system features a 2 × 2 monolithic array of 5-mm-pitch SDDs cooled with a double Peltier scheme and coupled to a four-channel CUBE charge preamplifier. Different digital pulse processors allowing operation in Mcps per second count rates are employed. The results of preliminary characterization measurements performed at both the LNF DAΦNE-Light DXR1 beamline and the ESRF LISA BM-08 are reported, in particular, XRF measurements on low atomic number elements (down to the Carbon K-line, 270 eV) and extended X-ray absorption fine structure of trace materials in pyrite.

## 1 | INTRODUCTION

Synchrotron beamlines are experiencing an increase of brilliance that demands an improvement in terms of count rate capability of existing detection systems, thus leading to the development of novel instrumentation.[1] Among different synchrotrons, the DAΦNE-Light DXR1 soft X-ray beamline at Frascati National Laboratory (LNF; Rome) is characterized for instance by a high-flux X-ray beam in the 1- to 3-keV energy range.[2] In the DXR1 beamline, X-ray absorption fine structure (XAFS) spectroscopy, a widely used analytical technique, is carried out on samples containing low-Z atomic elements. XAFS can be performed in both transmission and fluorescence mode. However, the transmission mode is restricted to samples characterized by the following specific characteristics: uniformity, thickness equal to ~2 times the absorption length, and absence of pinholes. For thick samples or lower concentrations, fluorescence mode XAFS is used.[3] In the XAFS fluorescence technique, a wide solid angle of the detector is essential, in order to collect most of the fluorescence isotropic emissions produced in the region where the scattering is minimum, improving the counting statistics of the spectrum.[4] For this method to be effective, the detector must have enough energy resolution to discriminate among different energy peaks, entailing a target energy resolution of 150-eV full width at half maximum (FWHM) Mn Kα line at 5.9 keV with a peaking time below 200 ns. Moreover, high throughput capability gives the possibility to perform the measurement with better counting statistics or faster acquisition time, taking

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advantage of the high flux of synchrotron light. Starting from these demands, in this work, we have attempted to deliver all the aforementioned requirements, aiming at overcoming the high-performance spectrometers based on silicon drift detectors (SDDs) for synchrotron applications at high count rates.

ARDESIA is a four-channel instrument prototype whose performance has been tested in two different synchrotron facilities: the LNF DAΦNE-Light DXR1 soft X-ray beamline and the ESRF LISA BM-08 beamline. In this work, the complete spectrometer is presented together with the preliminary results of its characterization carried out in those beamlines.

## 2 | THE ARDESIA SPECTROMETER

The spectrometer has been engineered in a finger-like structure, as shown in Figure 1, which allows the detector to be placed very close to the analyzed sample in an experimental chamber (as closest as ~20-mm total distance between SDD and the sample), thus covering a wide solid angle maximizing the photon collection. The mechanical structure of ARDESIA instrument is characterized by a base unit, a cold finger unit, the SDD detection module, and the external stainless steel covering tube. The first one has a parallelepiped shape, made in aluminum, and integrates a water-cooled heatsink and all the system connectors to external units. In the inner part of the spectrometer, under vacuum, the cold finger acts as mechanical support and heat transfer mean for the detector module screwed at its end. The external tube has a diameter of 38 mm and protects the internal parts of the spectrometer and maintains an internal static vacuum ($10^{-2}$ mbar). The vacuum is necessary to prevent ice formation, while the detector is cooled, and to maintain the vacuum environment in the sample chamber where the spectrometer is mounted. The cooling strategy is based on a water-cooled heatsink kept at 10°C by the chiller and two Peltier cells inside the instrument; this system is able to cool the detector down to −40°C. At the tip of the covering tube, the entrance window is found; it can be made of AP5 polymer,[5] graphite, or beryllium, depending on the energy range of operation in the specific synchrotron beamline. Beside the spectrometer, two external electronic systems manage the power supply and the cooling process of the instrument, making the ARDESIA spectrometer a standalone and easily transportable system.

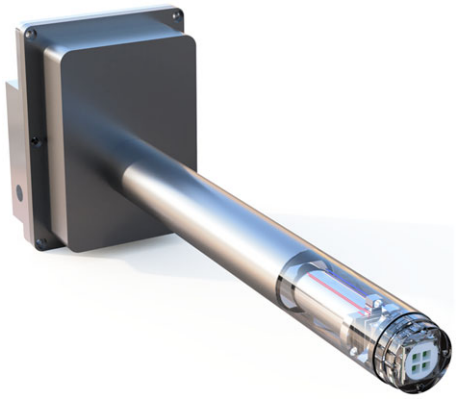


**FIGURE 1** Rendering of the ARDESIA spectrometer with semitransparent head showing the detection module with the four-pixel collimator. The vacuum-tight nose length is 25 cm, and the diameter is 38 mm

### 2.1 | Detection module

The four-channel detection module is composed by a monolithic 450-μm-thick 2 × 2 SDD matrix. SDDs have been chosen for their full sideward depletion and small anode capacitance, independent from the active area,[6] making them the ideal detector for high count rate applications. The SDDs mounted in the ARDESIA spectrometer are produced by Fondazione Bruno Kessler (Trento, Italy) using a low-leakage manufacturing process. Detectors are available in two different pixel shapes: 5-mm-pitch square and circular (25 and 20 $mm^2$ per pixel, collimated to 16 and 12.56 $mm^2$, respectively). The measurements here reported refer to the square geometry. The current signals from the detector anodes are integrated by a four-channel integrated preamplifier, CUBE,[7] directly wire bonded to the monolithic SDD matrix obtaining a very compact (16 mm × 16 mm) module.[8] Then the outputs of the preamplifier are processed by additional internal electronics for further amplification and reset generation. A complete characterization of the ARDESIA detection module has been made[8] reporting the energy resolution in terms of FWHM Mn Kα line of 125.7 eV at the optimum $t_{peak}$ = 3.2 μs and 151.5 eV at the $t_{peak}$ = 128 ns and $t_{flat}$ = 100 ns. In order to avoid the generation of charge between the boundary of different pixels, a collimator is placed over the detector. The collimator is made of Delrin® for energies up to 6 keV or of Mo for energies up to 20 keV.

### 2.2 | Auxiliary electronics

As anticipated, the ARDESIA spectrometer is accompanied by two dedicated external units: TESLA for the biasing of the internal electronics and the SDD detector and KRAKATOA that manages the instrument's thermal system. TESLA is a low-noise and disturbance-rejecting power supply unit, specifically designed to work in harsh

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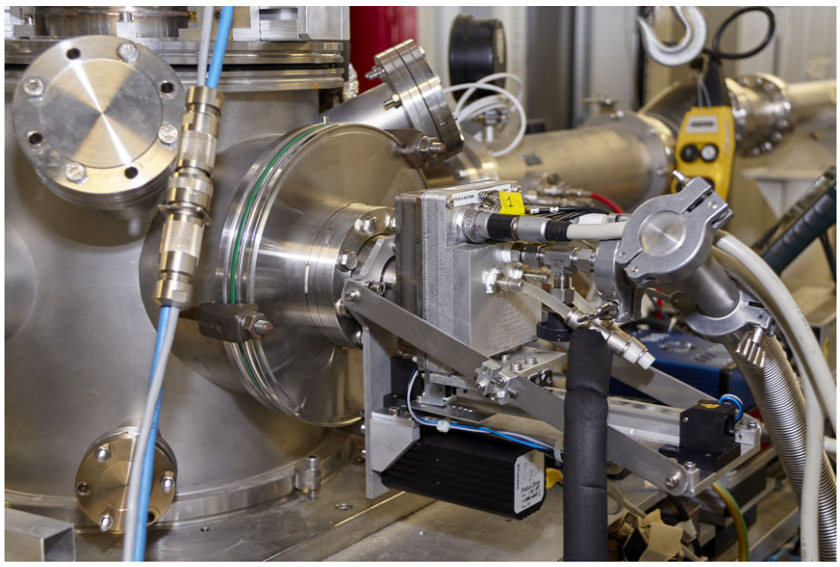


**FIGURE 2** Complete ARDESIA spectrometer system installed in LISA BM-08 ESRF beamline. Electrical connections and pipes for water cooling of the base Peltier element are visible

(from the point of view of electromagnetic interference's) environments, such as synchrotrons, providing clean supplies, current monitoring, and power sequencing features. KRAKATOA provides a closed-loop control of the two Peltier cells (thermo-electric cooling (TEC) modules) based on a firmware algorithm and the reading of four temperature sensors distributed in different points inside the spectrometer. The two units are powered by a compact 24-V AC-DC converter and provide USB communication to the beamline control room PC to monitor the system and completely automate the powering and cooling process with a user-friendly graphical user interface.

The spectrometer is mounted on a moving slide powered by a stepper motor to change the distance between the detector's head and the sample inside the scattering chamber. The vacuum inside the chamber is preserved by means of a double O-ring integrated with the interface flange. The complete ARDESIA system installed in the LISA BM-08 beamline is shown in Figure 2.

## 3 | EXPERIMENTAL RESULTS

The characterization campaigns of the ARDESIA instrument were carried out both at the LNF DAΦNE-Light DXR1 soft X-ray beamline and at the ESRF LISA BM-08 beamline.[9] The target of the measurements is to assess the detector performance in terms of energy resolution, high throughput capability, and stability over time, exploiting the synchrotron radiation and taking into account the electrically harsh environment of the beamline.

In the DAΦNE-Light DXR1 beamline, the energy resolution of the instrument was demonstrated by the acquisition of fluorescence spectrum from a common graphite sheet. Besides C (carbon), the sample also contains trace elements such as O, Na, Al, Si, P, and S. A four-channel digital pulse processor (XIA DXP-XMAP[10]) was used to process events with $t_{peak}$ = 2 μs and $t_{flat}$ = 100 ns, chosen for the optimum energy resolution. Figure 3 shows that the instrument is able to resolve the peaks in the spectrum from the fluorescence emission of the low-Z elements, as low as C (Z = 6, Kα1 at 277 eV).

In ESRF LISA BM-08 beamline, benefiting from a higher radiation intensity than the previous beamline, the performance assessment in terms of throughput capability was demonstrated through XRF measurements of Cu standard sample. During the measurements, two digital pulse processor (DPPs) were employed simultaneously: one-channel DANTE-4551[11] and four-channel XIA DXP-2X.[12] The input count rate (ICR) is modified by controlling the distance between sample and detector window. Different peaking times were chosen: $t_{peak}$ = 125 ns, which is the XIA's fastest shaping time,

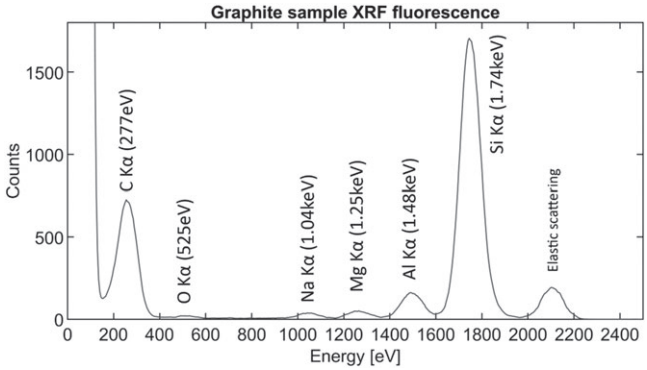



**FIGURE 3** X-ray fluorescence (XRF) measurement on a graphite sheet with an excitation energy of 2,100 eV performed at the DXR1 beamline

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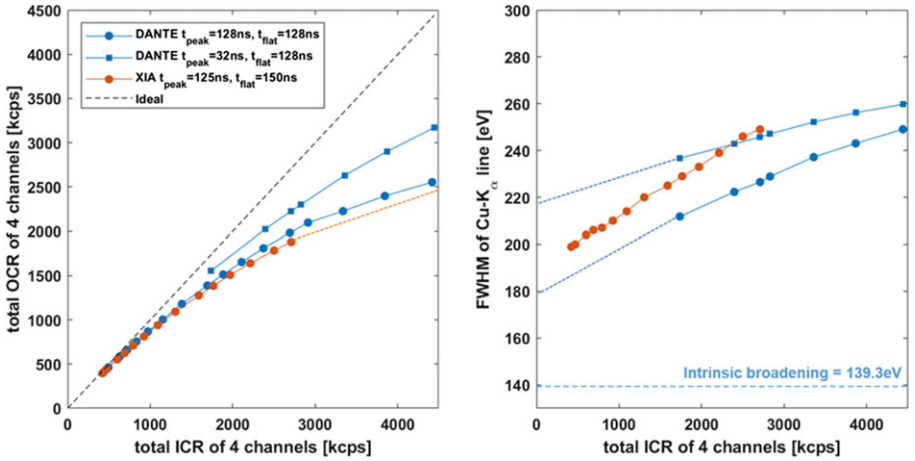



**FIGURE 4** Measured total OCR of four channels during X-ray fluorescence measurements with a Cu sample (left), and its associated energy resolution (full width at half maximum [FWHM]) as a function of input count rate (ICR; right), performed at the LISA BM-08 beamline

and $t_{peak}$ = 128 and 32 ns, with DANTE, being 32 ns the fastest available peaking time. As shown in Figure 4 (left), in the case of the highest ICR = 4.44 Mcps, an output count rate (OCR) = 2.55 and 3.17 Mcps of four channels have been acquired using $t_{peak}$ = 125 ns of XIA and 32 ns of DANTE, respectively. In terms of energy resolution, at the same ICR, the measured FWHM Cu Kα is 249.1 and 259.8 eV, respectively, for XIA and DANTE. The results confirm the high throughput capability of ARDESIA while still maintaining satisfactory energy resolution, although the significant increase of energy resolution with respect to the ICR still needs further investigation.

Experiments concerning absorption edge spectroscopy were also performed in the beamline. One of them was XAFS on pyrite ($FeS_2$), a common sulfide material. The experiment objective was to obtain the trace element's absorption edge in the sample, which requires very good energy resolution and stability of the instrument. Pyrite can include trace elements, such as As, Pb, and Tl. From this material, Tl $L_{III}$ edge absorption spectrum can be acquired using two possible regions of interest (ROIs) corresponding to the Tl Lα + As Kα lines and Tl Lα line, respectively, as shown in Figure 5 (left). Because the concentrations of S and Fe are dominant, a 50-mm Al filter was applied to partially reduce the emissions so that it will not occupy most of the detector processing times. However, it is still a challenging task to resolve the trace's characteristic lines, because the observed peak-to-peak ratio between Tl Lα and Fe Kα lines was still below 1%. The XAFS measurements were also repeated several times to check the stability of the instrument. The same XIA DXP-2X DPP is employed, using $t_{peak}$ = 500 ns and $t_{flat}$ = 150 ns for better observation of the Tl lines. Having ICR ≈ 20 kcps, $t_{acq}$ = 20 s per point, and total acquisition

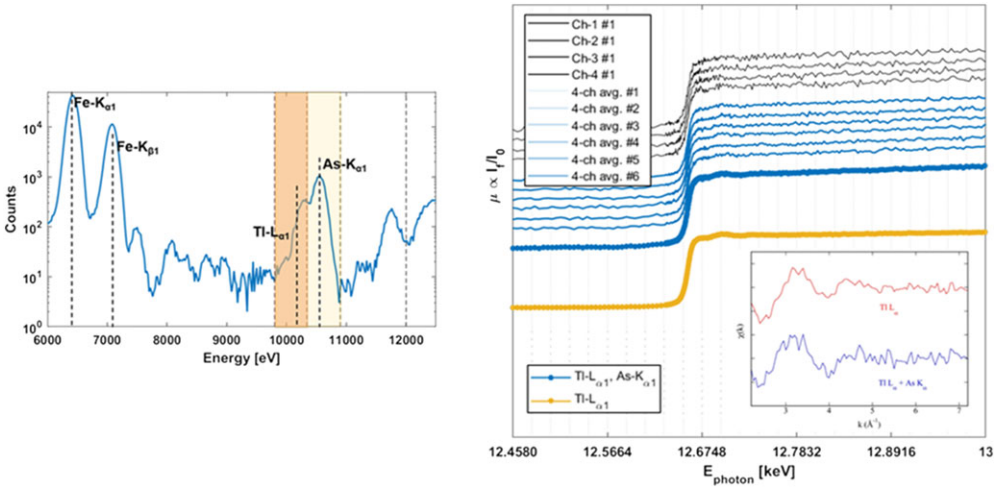



**FIGURE 5** Fluorescence spectrum of a pyrite sample (left) and its Tl $L_{III}$ edge absorption spectra (right), and extended X-ray absorption fine structure region transformed in *k*-space (right-inset), performed at the LISA BM-08 beamline

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time of 1.8 hr per measurement, the resulted extended XAFS (EXAFS) spectra are shown in Figure 5 (right). The spectra energy range is limited only to 13 keV due to the Pb Lα-Lβ lines, which are present after excitation energy above 13.05 keV, in the vicinity with Tl Lα and Lβ lines. Observing qualitatively the EXAFS spectra acquired on different measurement periods, no significant difference is present, validating the instrument stability. The method of repeating the measurements also helps reducing standard deviations of the EXAFS spectrum background. The comparison between spectra acquired by two different ROIs, indicated by the lower noise level of the EXAFS spectrum background, shows that the spectrum formed by Tl Lα ROI has better quality than the other spectrum.

## 4 | CONCLUSION

The complete ARDESIA spectrometer has been presented. Its preliminary characterization in two synchrotron beamlines has been reported, and it demonstrates the instrument's high performance: ~Mcps per second per channel count rate operation while maintaining satisfactory energy resolution. This is achieved thanks to the careful engineering of the compact detection module and low-noise design of the internal electronics and auxiliary modules. The modularity paves the way for further developments using thicker substrates and larger arrays.

## ACKNOWLEDGEMENTS

The project is supported by the Italian INFN. The authors thankfully acknowledge Francesco D'Acapito from ESRF.

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**How to cite this article:** Hafizh I, Bellotti G, Carminati M, et al. ARDESIA: A fast silicon drift detector X-ray spectrometer for synchrotron applications. *X-Ray Spectrometry*. 2019;48:382–386. https://doi.org/10.1002/xrs.3017

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